

Course Code- 151610 Internet & Intranet Engineering**3 0 0 3****Unit 1: Fundamentals of Internet & Intranet**

Introduction to Internet and Intranet, Evolution of networking, Internet architecture and standards, Intranet applications in enterprises, TCP/IP model, IPv4 vs IPv6, Domain Name System (DNS), HTTP/HTTPS protocols.

Unit 2: Web Technologies and Communication Protocols

Web architecture, Client-server model, HTTP and Restful APIs, Web services (SOAP, REST), FTP, SMTP, IMAP, WebSockets, Secure communication protocols (TLS, SSL), Content Delivery Networks (CDN).

Unit 3: Intranet Design & Enterprise Networking

Intranet components, Network design principles, VLANs and VPNs for intranet, Authentication and access control, Directory services (LDAP, Active Directory), Single Sign-On (SSO), Proxy servers and load balancing.

Unit 4: Cloud Computing and Virtualization

Introduction to cloud computing, Intranet integration with cloud services, Virtualization technologies (VMware, Hyper-V, KVM), Software-defined networking (SDN), Edge computing for intranet solutions, Serverless architecture.

Unit 5: Security and Performance Optimization

Security threats in intranet and internet environments, Firewalls and IDS/IPS, Zero Trust Security model, Performance tuning and caching strategies, Content optimization (compression, minification, lazy loading), Secure API development.

Unit 6: Emerging Trends and Case Studies

IoT and Industrial Intranet, Blockchain for secure communication, AI-driven networking, Digital Twins for enterprise intranets, Case studies on large-scale intranet implementations, Future trends in Internet and Intranet technologies.

Text/Reference:

1. Douglas E. Comer, Internetworking with TCP/IP, Pearson.
2. Behrouz A. Forouzan, Data Communications and Networking, McGraw Hill.
3. Andrew S. Tanenbaum, Computer Networks, Pearson.
4. William Stallings, Data and Computer Communications, Pearson.
5. Randy J. Boyle, Internet & Intranet Security, Wiley.
6. Kai Hwang, Geoffrey Fox, Jack Dongarra, Distributed and Cloud Computing, Elsevier.
7. Paul Deitel, Harvey Deitel, Internet & World Wide Web: How to Program, Pearson.